

# Lecture 9

## Sampling, Experiments and Causation

# Review

## **1 What is bias?**

When a sampling method's estimates miss the parameter in the same direction every time - an accuracy problem.

## **2 What is standard error?**

How much an estimate varies from one sample to the next - a precision problem.

## **3 What is random sampling?**

Using chance, not a person, to decide which units get into the sample.

## **4 What is a simple random sample (SRS)?**

Every possible group of  $n$  units has the same chance of being the sample.

# Review

## **5 What is a convenience sample?**

A sample of whoever is easiest to reach.

## **6 What is sampling with replacement?**

A picked unit goes back into the pool, so it can be picked again.

## **7 What is sampling without replacement?**

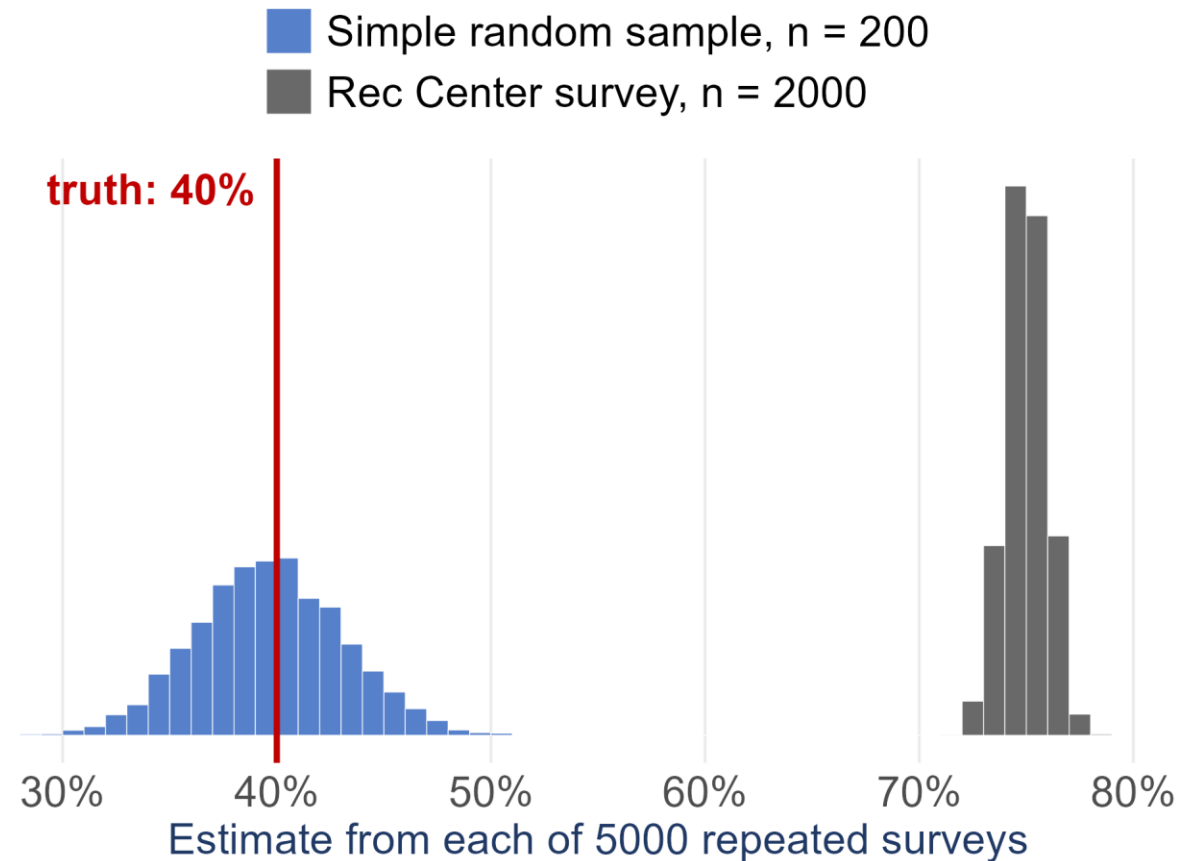
A picked unit is set aside, so it can be chosen only once.

## **8 What is a voluntary response sample?**

A sample of people who choose themselves by responding - usually the ones with strong opinions.

# A bigger sample does not fix bias

- Suppose 40% of UI students exercise at least three times a week
- Blue: 5000 simple random samples of 200 students
- Grey: 5000 Rec Center surveys of 2000 students
- The Rec Center survey is more **precise** - a smaller standard error
- and every one of its estimates misses the truth: **bias**
- More data buys precision. Only a **better design** buys accuracy



# In the news: how a national poll samples

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- State of the Facts 2026 (AP-NORC Center with USAFacts) asked U.S. adults how much they trust government information and AI
- 1,036 adults, July 29 – August 10, 2026: 1,017 online and 19 by phone, in English or Spanish
- Drawn from AmeriSpeak, a **probability-based panel**: NORC randomly selects U.S. addresses, then recruits those households by mail, phone and in person
- You cannot sign yourself up - **chance decides** who is invited. Compare the opt-in panels from last week
- Margin of sampling error:  $\pm 4.1$  percentage points
- But only **10.3%** of invited panel members completed this survey (AAPOR response rate 3)
- One finding: only 37% say it is easy to know whether the information they see is true

Read more: [Project page](#) · [Full report \(PDF; methods on p. 42\)](#) · [How AmeriSpeak recruits](#)

*Sources: AP-NORC Center for Public Affairs Research and USAFacts (2026); NORC, About AmeriSpeak*

# In the news: light at night, over time

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- UK Biobank: about 500,000 UK adults aged 40–69 who agreed to join in 2006–2010, then followed through their health records
- 2013–2015: 103,615 of them wore a wrist sensor with a light meter for 7 days
- Removed: unreliable sensor data, 2,770 night-shift workers, anyone with heart disease already, anyone without a later heart MRI → 11,071 people
- About 3 years later each had an MRI of their heart; hospital and death records were followed for up to 10 years
- **Longitudinal**: the same people measured more than once - night light first, the heart later
- **Not random**: the people who agreed to join are mostly White and healthier than the UK population (the authors say so)
- Result: more light above 3 lux at night went with thicker heart walls and weaker pumping. Did the light cause it? We come back to that later today

Read more: [ESC press release](#) · [Dai et al. \(2026\), European Heart Journal](#)

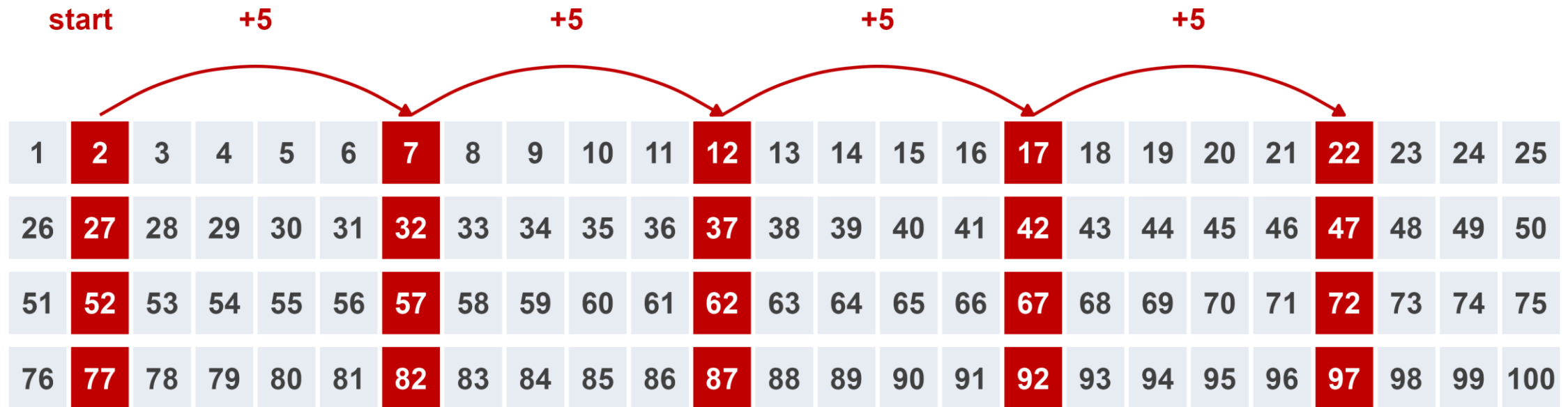
*Sources: Dai, Dai, Heianza and Qi (2026), European Heart Journal; ESC press release (2026)*

# Systematic sample

## systematic sample

pick a random starting point among the first  $k$  units of an ordered list, then take every  $k$ th unit after it

$k = N/n = 100/20 = 5$ . Random start = 2, so we take list positions 2, 7, 12, 17, ... , 97



Works like an SRS - unless the list has a repeating pattern that lines up with  $k$

# Stratified random sample

## strata

groups of units that are alike, formed BEFORE sampling - class year, region, age group

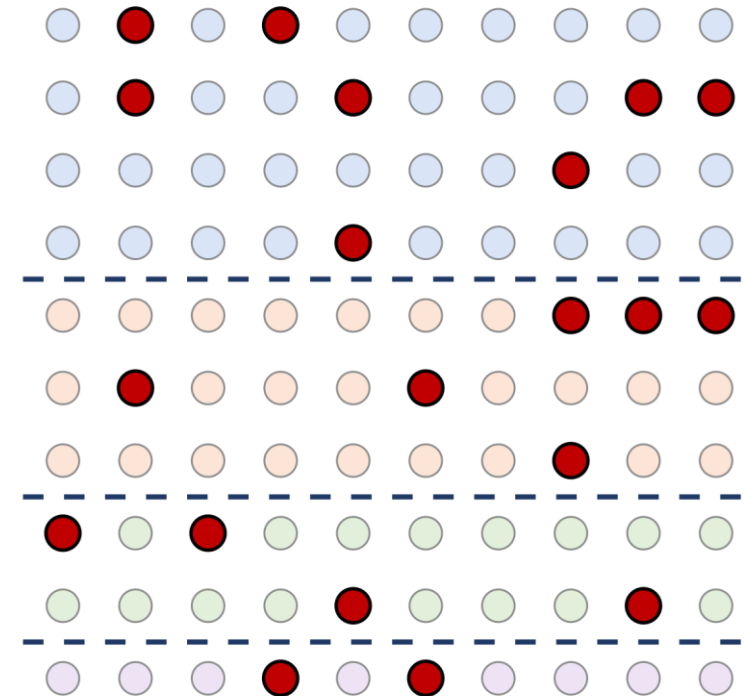
## stratified random sample

a separate SRS inside every stratum, combined into one sample

## proportional allocation

each stratum gets its share of  $n$ : 40, 30, 20, 10 students  $\rightarrow$  8, 6, 4, 2

**Every group is guaranteed a place - no more 2 juniors by bad luck**



● First-year    ● Junior  
● Sophomore   ● Senior

# Proportional allocation with real data

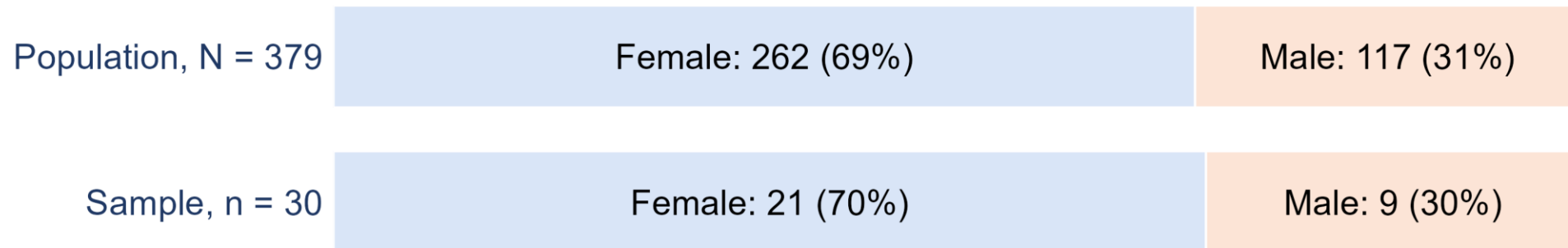
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heights.csv (on the course Data page) has  $N = 379$  people: 262 female and 117 male. Stratify by sex for a sample of  $n = 30$ :

female:  $30 \times 262/379 = 20.74$ , so 21

male:  $30 \times 117/379 = 9.26$ , so 9

Then an SRS of 21 of the women, and a separate SRS of 9 of the men



# Cluster sample

## cluster

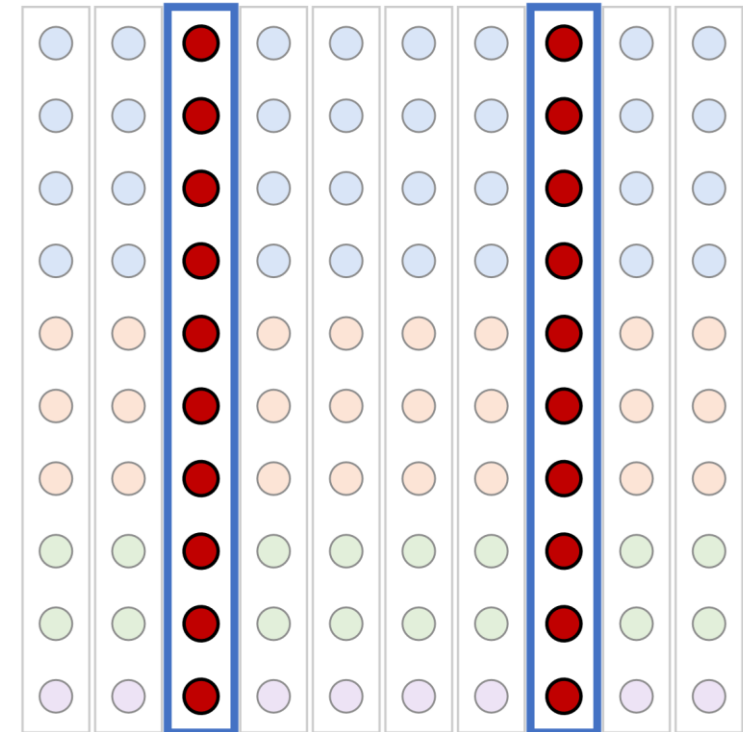
a group of units that occurs naturally - a dorm, a classroom, a city block

## cluster sample

an SRS of clusters, then measure EVERY unit in the clusters chosen

Only needs a list of clusters, not of every person - and far less travel

Works best when each cluster is a mini-population: mixed inside, like each column here



● First-year    ● Junior  
● Sophomore   ● Senior

# Two-stage cluster sample

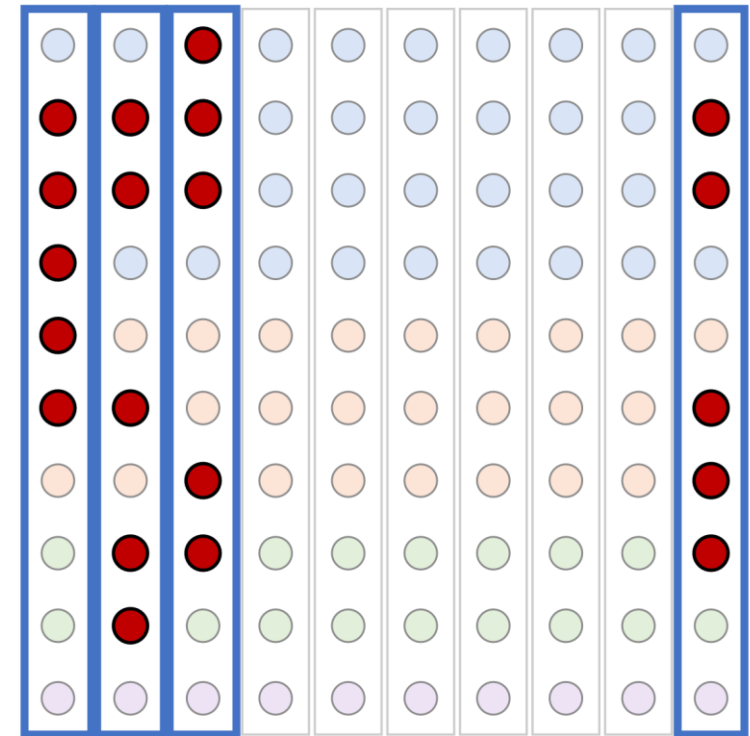
## two-stage cluster sample

stage 1: an SRS of clusters. Stage 2: an SRS of units inside each chosen cluster

Here: 4 of the 10 columns, then 5 students in each,  $n = 20$

Exit polls work this way: sample polling places, then interview every  $n$ th voter leaving each one

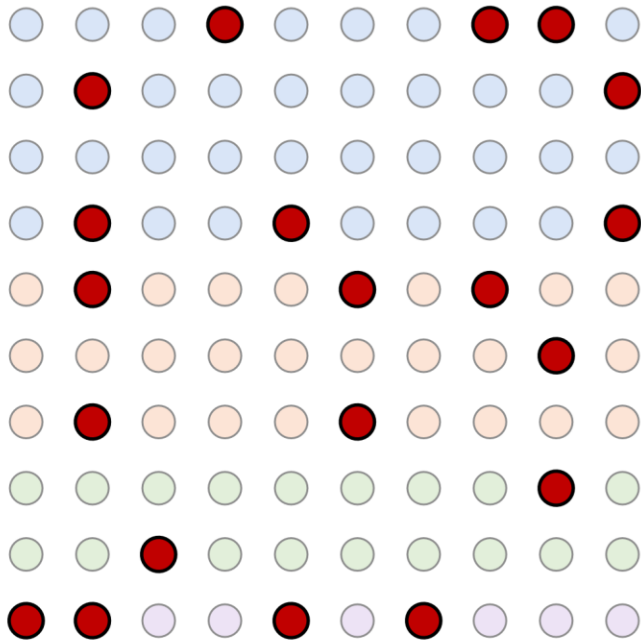
National surveys such as NHANES stack it all: strata, clusters inside strata, households inside clusters



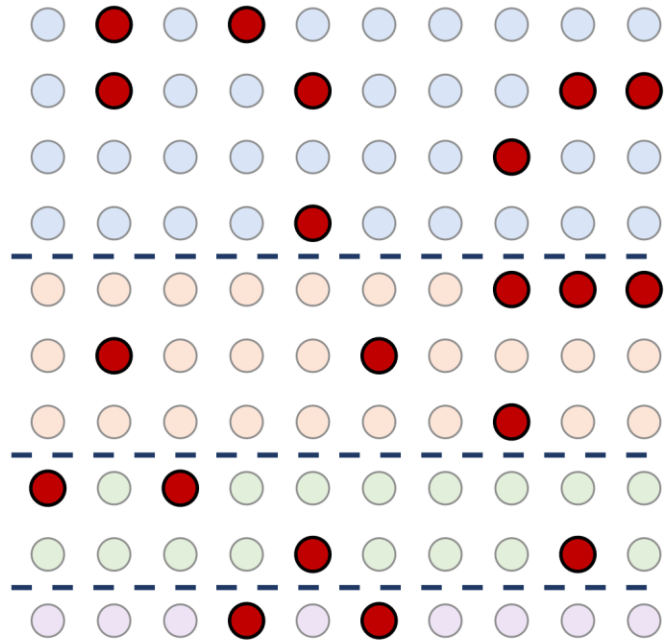
● First-year    ● Junior  
● Sophomore   ● Senior

# Strata vs. clusters

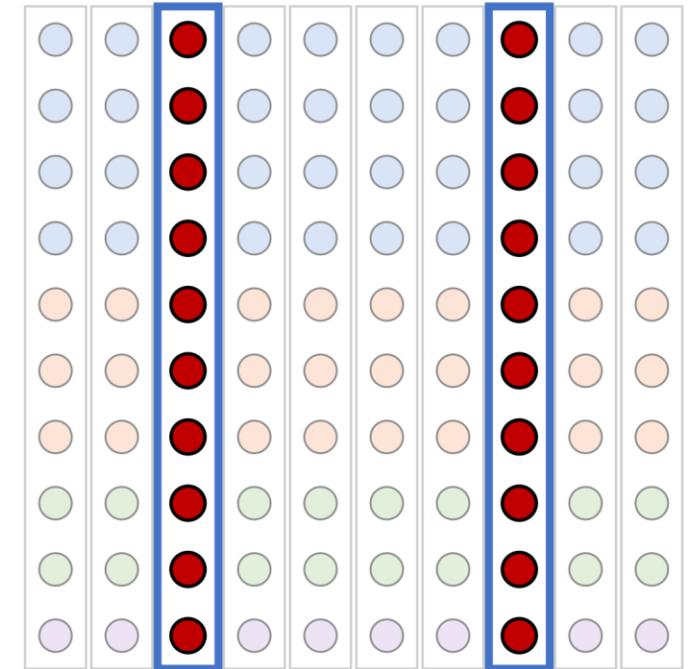
Simple random



Stratified



Cluster



● First-year ● Sophomore ● Junior ● Senior

**Strata: alike inside - sample from EVERY one.    Clusters: mixed inside - sample SOME, whole.**

# Two kinds of error

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## **sampling error**

the natural gap between a sample and the population, caused by which units happened to be chosen

*every sample has some - a larger sample makes it smaller*

## **nonsampling error**

errors that do not come from chance - biased methods, bad questions, data-entry mistakes, faulty instruments

*a larger sample does not make these smaller*

## **sampling bias**

some members of the population are less likely to be chosen than others

*convenience and voluntary response samples, or a frame that leaves people out*

# Three common nonsampling errors

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Even a perfectly random design can end up biased:

## **undercoverage**

the sampling frame leaves part of the population out  
*students who are not on the list you sampled from*

## **nonresponse**

people who are chosen cannot be reached, or refuse to answer  
*a problem when the people who do not answer differ from those who do*

## **response bias**

the answers themselves are wrong  
*leading or confusing questions, sensitive topics, wanting to please the interviewer*

# Nonresponse: when nobody picks up

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Pew Research Center reached 36% of the people it called for its phone surveys in 1997. By 2018 it reached 6%.

1. Does a 6% response rate automatically make a survey biased?

**No - only if the people who answer differ from the people who do not. The trouble is that you usually cannot check.**

2. In 2020, national polls in the final two weeks overstated Biden's margin by 3.9 points on average - the largest error in 40 years. What could cause that?

**The pollsters' own task force concluded that at least some of the error came from nonresponse: the people who answered differed from those who did not.**

*Sources: Pew Research Center (2019); AAPOR Task Force on 2020 Pre-Election Polling (2021)*

# Question wording: same law, two names

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In September 2013, CNBC asked half of its poll respondents about “Obamacare” and the other half about “the Affordable Care Act” - the same law.

1. Which name do you think drew more opposition?

**Obamacare: 46% opposed it, against 37% for the Affordable Care Act.**

2. So did fewer people support Obamacare, too?

**No - 29% supported Obamacare and 22% the ACA. 30% said they did not know about the ACA, but only 12% about Obamacare.**

3. What does this tell you about survey questions?

**The words in a question change the answers. A familiar, politically loaded name pulled people off the fence.**

*Sources: NBC News (September 27, 2013); Harvard Political Review*

# Practice: name the sampling design

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1. A reporter interviews the first 50 students who walk out of the Idaho Student Union Building.  
**Convenience sample - only students who happen to pass the SUB can be chosen.**
2. The registrar numbers every enrolled student, and a computer picks 200 of the numbers at random.  
**Simple random sample.**
3. A principal polls 50 first-years, 50 sophomores, 50 juniors and 50 seniors, chosen at random within each year.  
**Stratified random sample - the class years are the strata.**
4. 5 of the 40 apartment buildings in town are chosen at random, and every household in them is surveyed.  
**Cluster sample - the buildings are the clusters.**
5. A medical researcher interviews every third patient on a hospital list, starting from a randomly chosen one of the first three.  
**Systematic sample, with  $k = 3$ .**

# Your course project: Part 3

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- **Part 3** asks you to draw a sample of 20 to 30 observations from your dataset
- Pick a design: simple random, systematic, stratified or cluster sampling
- Say exactly how you drew it - the method, n, and the tool - so someone else could repeat it
- Justify the choice for YOUR data: is there a natural group to stratify on? natural clusters?
- Name any bias your method could introduce
- **Steps 1 to 4** of the project can be started as soon as this lecture is finished

# Part 2

## Experiments and Causation

# Observational studies and experiments

## **response variable**

the outcome we measure  
*heart disease*

## **explanatory variable**

the variable we think explains or influences the response  
*hormone therapy (yes or no)*

## **observational study**

observe individuals and measure variables, without influencing anything  
*survey women about the hormones they chose to take*

## **experiment**

the researcher deliberately assigns treatments, then measures the response  
*randomly assign women to hormones or a placebo*

Ask: who decided which group each unit ended up in? The units themselves -> observational. The researcher -> experiment.

# Association is not causation

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## **association**

two variables move together in the data

## **causation**

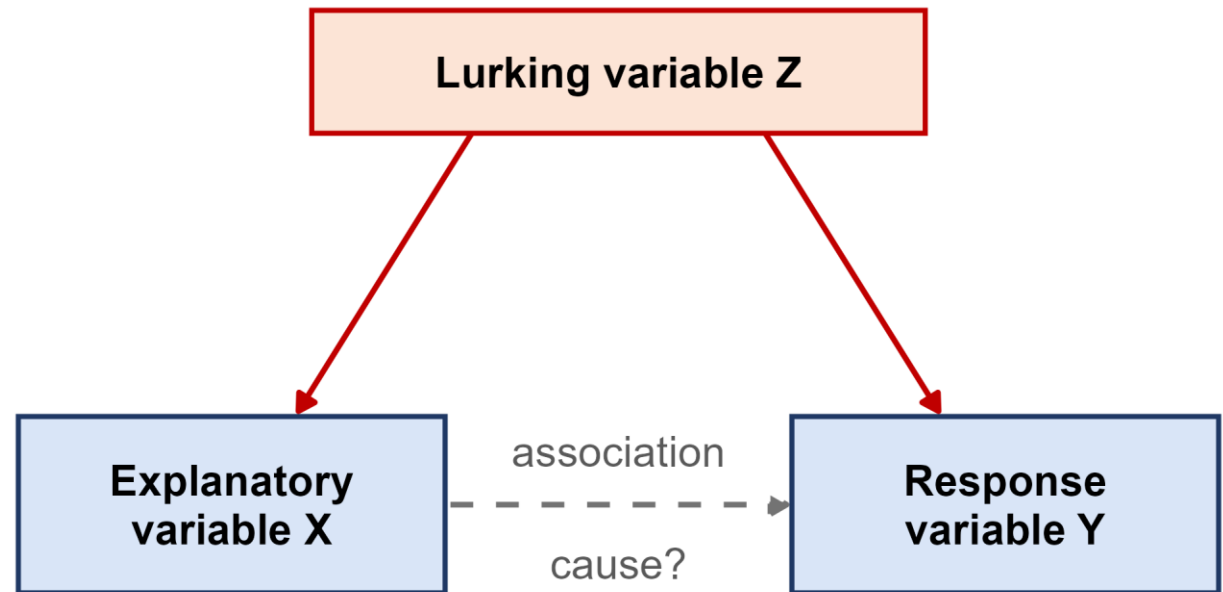
changing one variable would change the other

## **lurking variable**

a variable not in the study that influences both the explanatory and the response variable

## **confounded**

two variables whose effects on the response cannot be separated



Ice cream sales and drownings rise together. The lurking variable: hot weather. The association is real - the causal story is not.

# Hormone therapy and heart disease

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Large observational studies, such as the Nurses' Health Study, found that women who took hormone therapy after menopause had less heart disease

The Women's Health Initiative then randomly assigned 16,608 women aged 50 to 79 to estrogen plus progestin or a placebo

It was stopped early in 2002 because the risks outweighed the benefits: more breast cancer, and MORE heart disease, not less

**Why did the observational studies point the wrong way?**

## **Lurking variables.**

Women who chose hormone therapy tended to be wealthier, better educated and healthier to start with - and that, not the hormones, lowered their heart risk. Random assignment balanced those differences out.

*Source: Writing Group for the WHI Investigators (2002), JAMA*

# Coffee and the risk of death

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The NIH-AARP study followed 402,260 adults aged 50 to 71 from 1995 to 2008. Before adjusting for anything but age, men who drank 6+ cups of coffee a day had a 60% HIGHER risk of death than men who drank none

But 35% of those heavy coffee drinkers smoked, compared with 5% of the men who drank no coffee

After adjusting for smoking and other factors, the same men had a 10% LOWER risk of death - and fewer deaths from heart disease

**Why did the coffee effect flip?**

**Lurking variable: smoking.**

Heavy coffee drinkers smoked far more, and smoking - not coffee - raised their risk. Even after adjusting, the authors say they cannot tell whether coffee itself helps.

*Source: Freedman et al. (2012), New England Journal of Medicine*

# Can we claim cause without an experiment?

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- Nobody can randomly assign people to smoke for thirty years - or randomly assign their birth order
- Yet we are confident that smoking causes lung cancer. Decades of observational studies found:
  - a strong association, found again and again in different populations
  - a dose-response pattern: the more cigarettes, the higher the risk
  - the right order in time: the smoking comes before the cancer
  - a plausible biological explanation, backed by laboratory evidence
- The 2014 Surgeon General's report: smoking causes at least 480,000 premature U.S. deaths a year, about 127,700 of them from lung cancer
- Without an experiment, a causal claim needs this much evidence - **never one study**

*Source: U.S. Department of Health and Human Services (2014)*

# Back to light at night and the heart

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Recall the UK Biobank study from slide 6: 11,071 adults wore a wrist light sensor for 7 days in 2013-2015, then had a heart MRI about 3 years later.

1. Is this an observational study or an experiment?

**Observational. Nobody was assigned a bright or a dark bedroom - the light people already had was measured.**

2. People with the most light above 3 lux had 1.5% thicker heart walls than people with none. Did the light cause it?

**This study cannot tell us. It shows an association - the authors say their design cannot establish cause.**

3. The press release was titled “Too much light at night changes the shape and working of the heart.” Fair?

**It goes too far. The paper says more night light was associated with these changes, not that it changed the heart.**

*Sources: Dai et al. (2026) and Münzel, Kuntic and Daiber (2026), European Heart Journal; ESC press release (2026)*

# Light at night: how strong is the evidence?

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The authors adjusted for age, smoking, physical activity, BMI, air pollution, deprivation and more. The heart changes grew steadily as night light went up.

1. They adjusted for many variables. Is that as good as random assignment?

**No. Adjusting only handles what was measured. Traffic noise was not, and the editorial says it could confound.**

2. Which features from the smoking evidence does this study have? What is it missing?

**Time order, a dose-response and a plausible link (shorter sleep). Missing: many studies finding it again.**

3. The participants were volunteers, mostly White and healthier than average. Who can we generalize to?

**Only people like them. Volunteers are not a random sample of any population.**

*Sources: Dai et al. (2026) and Münzel, Kuntic and Daiber (2026), European Heart Journal*

# The language of experiments

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**experimental units**

the individuals or objects the treatments are applied to

**subjects**

experimental units that are people

**factor**

an explanatory variable the researcher controls

**levels**

the specific values a factor is set to

**treatment**

the condition a unit receives: one combination of factor levels

# Example: the VITAL trial

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VITAL randomly assigned 25,871 U.S. adults to vitamin D (2,000 IU a day) or a placebo, AND to omega-3 fish oil (1 g a day) or a placebo, then followed them for about 5 years.

1. What are the experimental units - and what do we call them?

**The 25,871 adults. They are people, so they are subjects.**

2. What are the factors, and how many levels does each have?

**Vitamin D and omega-3 - 2 levels each (the supplement or a placebo).**

3. How many treatments are there?

**$2 \times 2 = 4$ : vitamin D + omega-3, vitamin D + placebo, placebo + omega-3, placebo + placebo.**

4. What happened?

**Neither supplement lowered the rate of invasive cancer or of major cardiovascular events.**

*Source: Manson et al. (2019), New England Journal of Medicine*

# Principles of experimental design

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**control**

compare the treatment against a control group, so the treatment is the only systematic difference between the groups

**random assignment**

let chance assign units to treatments - it balances the lurking variables we know about AND the ones we do not

**replication**

put enough units in each group that chance differences between the groups average out

# Placebos and blinding

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**placebo**

an inactive treatment that looks exactly like the real one  
*a sugar pill, or a saline injection*

**placebo effect**

people often improve simply because they believe they are being treated  
*in one study, believing you had taken a performance drug worked almost as well as the drug itself*

**blinding**

the subjects do not know which treatment they received

**double-blind**

neither the subjects nor the researchers working with them know who received what

# The 2020 Pfizer-BioNTech vaccine trial

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43,548 people were randomly assigned, half and half, to two shots of the COVID-19 vaccine or two shots of a saline placebo. Participants and the staff assessing them did not know who got which. Afterwards: 8 COVID-19 cases in the vaccine group, 162 in the placebo group.

1. Why give the control group a saline shot instead of no shot at all?

**So both groups have the same experience. A difference in cases can then only come from the vaccine, not from expectations.**

2. Why randomize, instead of letting people choose whether to get the vaccine?

**People who choose a new vaccine may also be more careful in other ways - a lurking variable. Randomizing balances it.**

3. What does 8 cases versus 162 suggest?

**The vaccine group had about 95% fewer cases - and because of the design, we can credit that to the vaccine.**

*Source: Polack et al. (2020), New England Journal of Medicine*

# A placebo surgery

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Arthroscopic knee surgery for arthritis was common, and patients said it helped

Moseley and colleagues (2002) randomly assigned 180 patients with knee osteoarthritis to two real arthroscopic procedures, or to placebo surgery - skin incisions only, with no instruments inserted

Patients, and the staff who assessed their pain, did not know who got which

**Over two years, which group reported the most improvement?**

**None of them.**

The placebo group did as well as both surgery groups on pain and function. The benefit came from believing you had surgery - and without a placebo group and blinding, nobody would have known.

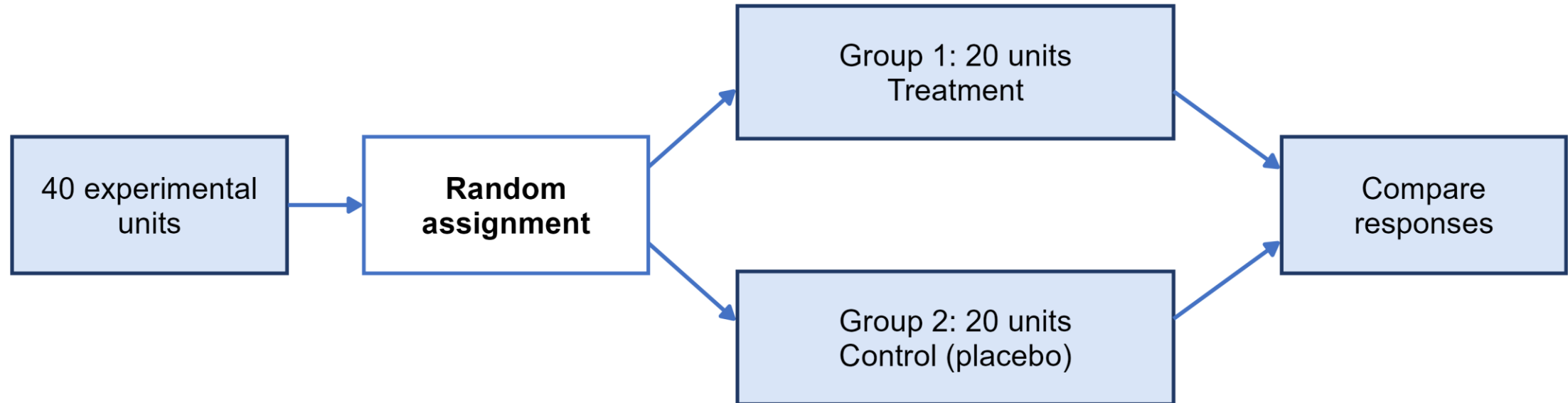
*Source: Moseley et al. (2002), New England Journal of Medicine*

# Completely randomized design

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## completely randomized design

every unit is assigned to a treatment completely at random - the simplest design



The Pfizer-BioNTech vaccine trial was a completely randomized design with two groups

# Randomized block design

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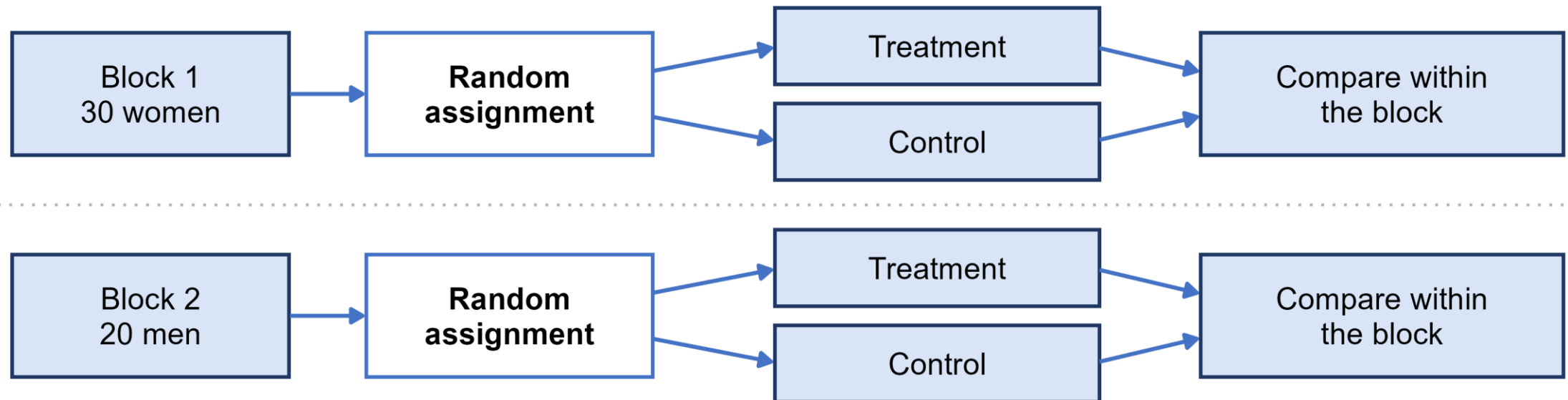
## block

a group of units that are alike in a way that affects the response, formed BEFORE treatments are assigned

## randomized block design

randomize treatments separately inside each block, then compare treatments within blocks

*blocking is to experiments what stratifying is to sampling*

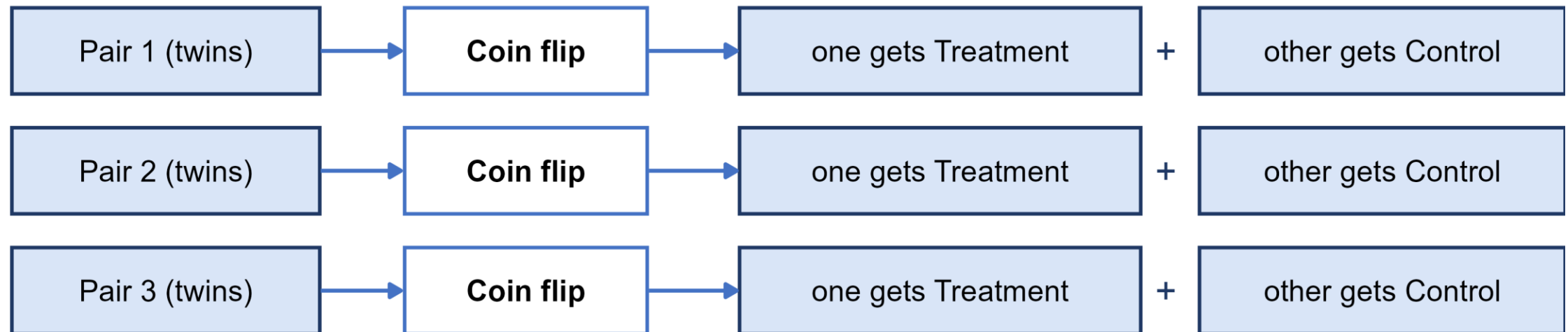


# Matched pairs design

## matched pairs design

a block design where every block is a pair of very similar units, and a coin flip decides which one gets each treatment

*or each subject is their own pair, receiving both treatments in a random order*



Before-and-after data with no random order or control group is much weaker evidence

# What does randomization buy you?

**Random sampling** lets you generalize from the sample to the population

**Random assignment** lets you conclude cause and effect

Most studies have only one. A trial on volunteers shows a vaccine works - for people like those volunteers

|                   |     | Randomly ASSIGNED to treatments?                                                                   |                                                                                             |
|-------------------|-----|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|                   |     | Yes                                                                                                | No                                                                                          |
| Randomly SAMPLED? | Yes | <b>Cause, AND generalize to the population</b><br><br>rare: an experiment on a random sample       | <b>Generalize, but not cause</b><br><br>e.g. a random-sample survey                         |
|                   | No  | <b>Cause, but only for units like these</b><br><br>e.g. most clinical trials, which use volunteers | <b>Association in these units only</b><br><br>e.g. volunteers who chose their own treatment |

# Practice: design the study

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You want to know whether texting slows how quickly a driver reacts when the car ahead brakes.

1. What are the explanatory variable, the response variable, and the treatments?

**Explanatory: texting or not. Response: reaction time. Treatments: driving while texting, and driving only.**

2. Your partner wants to split drivers at random into a texting group and a driving-only group. Is there a better design?

**Matched pairs: each driver does both, in a random order. Reaction times vary a lot between people, so compare each driver to themselves.**

3. Can the study be blinded?

**Not for the drivers - they know when they are texting. But the person measuring reaction times can be kept blind.**

*Adapted from Introductory Statistics 2e (OpenStax), Section 1.4, Try It 1.21*

# Ethics: the Facebook experiment

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For one week in January 2012, Facebook changed the News Feeds of 689,003 users: some saw fewer positive posts, some fewer negative ones. The results were published in 2014. The users never knew.

1. Facebook said agreeing to its Data Use Policy counted as informed consent. Do you agree?

**The journal did not: its Expression of Concern said the study may not have met the principles of informed consent and the chance to opt out.**

2. Why did no ethics board stop it?

**Facebook, a private company, was not bound by the federal rules. Cornell's board decided its researcher was not directly involved, so it did no review.**

3. What is the lesson?

**Clicking I agree on a terms-of-service page is not informed consent - even for a well-designed experiment.**

*Sources: Kramer, Guillory and Hancock (2014), PNAS; PNAS Editorial Expression of Concern (2014)*

# The Belmont Report: three principles

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*Why the rules exist: in the U.S. Public Health Service syphilis study at Tuskegee (1932 to 1972), hundreds of Black men with syphilis were left untreated without informed consent. The Belmont Report (1979) followed.*

## **respect for persons**

people decide for themselves whether to take part

## **beneficence**

do no harm: maximize the possible benefits and minimize the possible risks

## **justice**

the burdens and benefits of research are shared fairly

# Research ethics today

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## **informed consent**

subjects are told the purpose, procedures and risks in advance, and can refuse or quit at any time

## **confidentiality**

information about individuals is protected and never released in a form that identifies them

## **Institutional Review Board (IRB)**

reviews and approves studies on human subjects before they start  
*at UI, through the Office of Research Assurances*

# Ethics: honest data

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Ethics also means honest data. An investigation of psychologist Diederik Stapel found he had invented data: it tainted more than 55 of his papers and 10 PhD dissertations he supervised. A researcher is surveying households in a community:

1. They choose a block where they already know many of the neighbors. What is the problem?  
**A convenience sample - it is biased, and claiming it represents the community is misleading.**
2. Nobody is home at four houses, so they skip them and never go back. What is the problem?  
**Nonresponse - people who are not home, such as working families, may differ from the people who are.**
3. Running late, they fill in four forms with answers copied from neighbors. What is the problem?  
**Fabricated data. It is never acceptable, even when the answers look realistic.**

*Source: Introductory Statistics 2e (OpenStax), Section 1.4 and Example 1.22*

# Practice: what design, and can it show cause?

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1. A survey finds that people who own dogs have lower blood pressure than people who do not.  
**Observational - nobody was assigned a dog. Active people may both own dogs and have lower blood pressure.**
2. 60 volunteers with insomnia are randomly split: 30 get melatonin, 30 get an identical-looking placebo.  
**Completely randomized experiment - it can show cause, for people like these volunteers.**
3. In each of 4 greenhouses, tomato plants are randomly assigned to fertilizer A or fertilizer B.  
**Randomized block design, with greenhouses as the blocks - it can show cause.**
4. Subjects complete mazes wearing a floral-scented mask and an unscented mask, in a random order. Can the subjects be blinded?  
**No - they can smell the flowers. But the researchers timing the mazes can be blinded. (Textbook Example 1.20)**